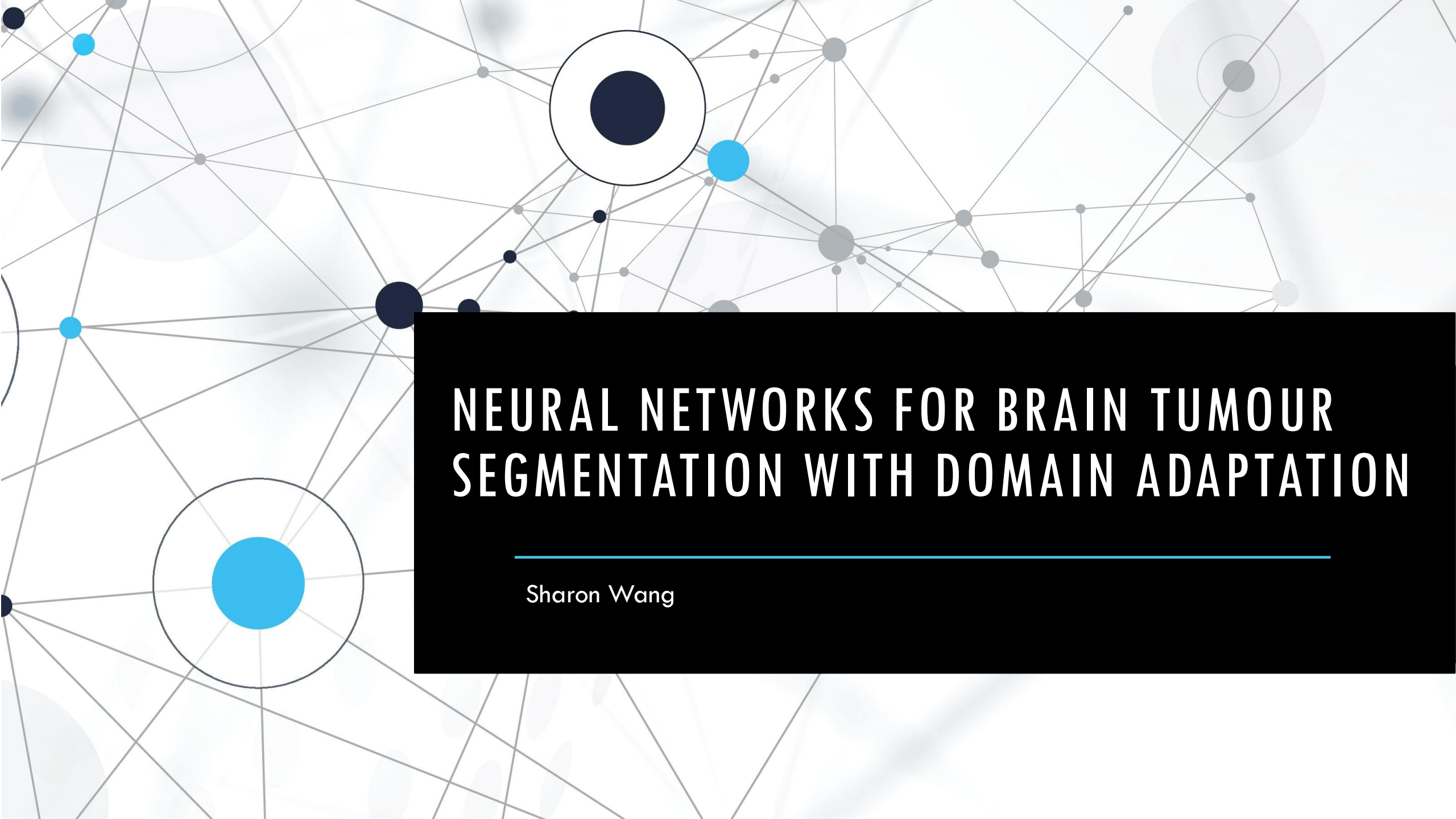
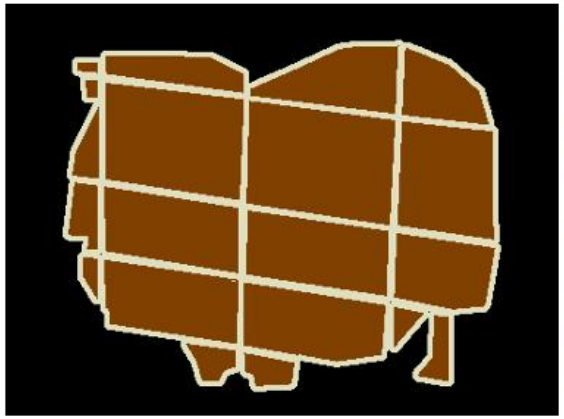


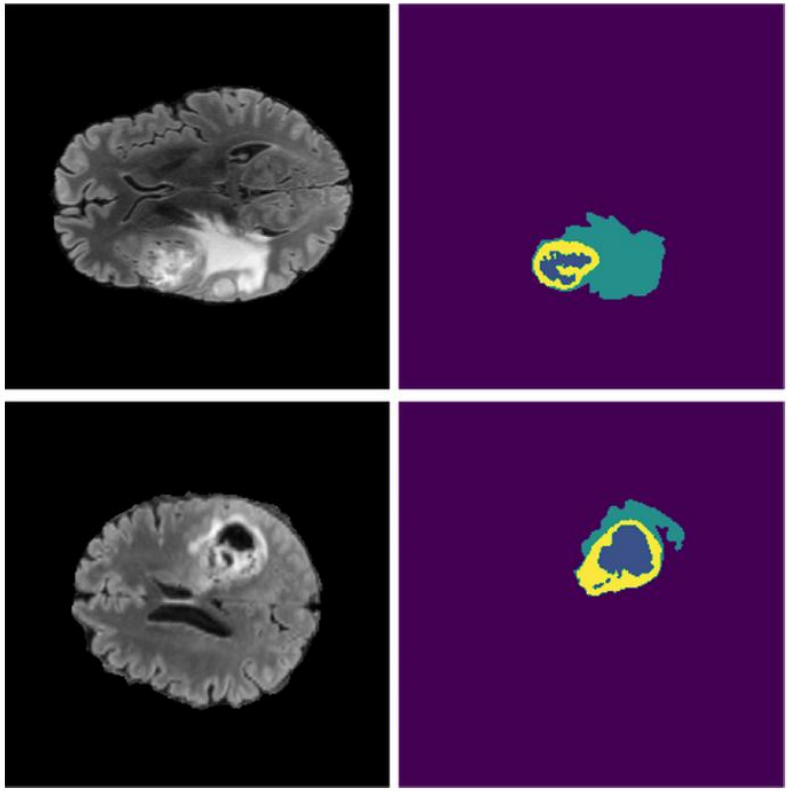
NEURAL NETWORKS FOR BRAIN TUMOUR SEGMENTATION WITH DOMAIN ADAPTATION

Sharon Wang

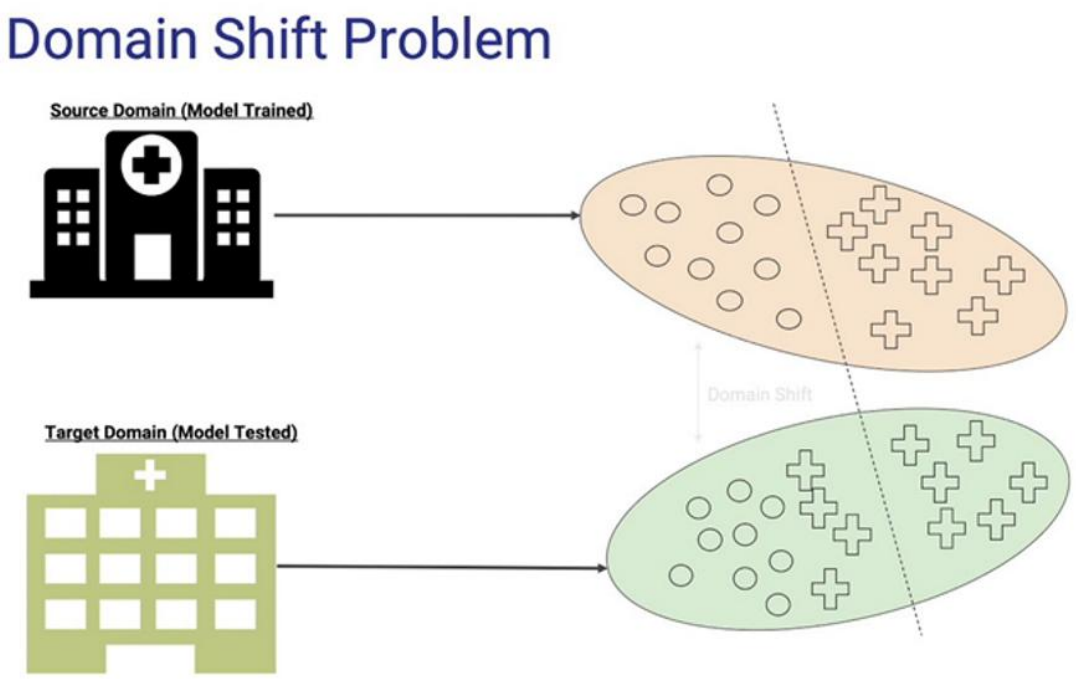
PROBLEM STATEMENT

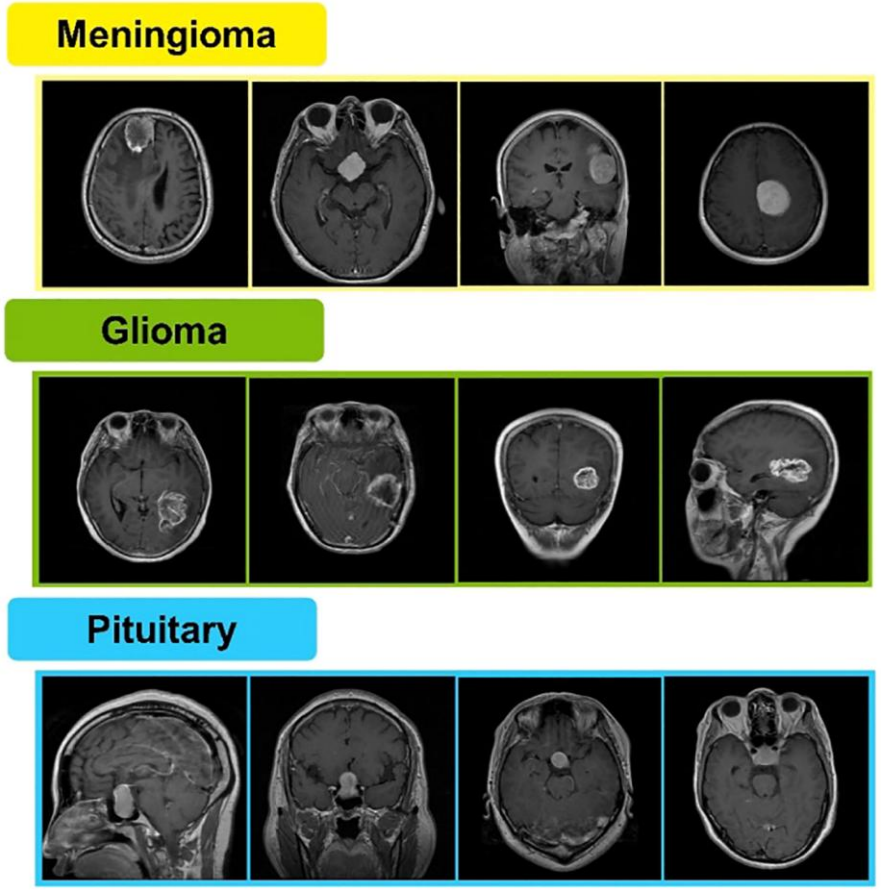




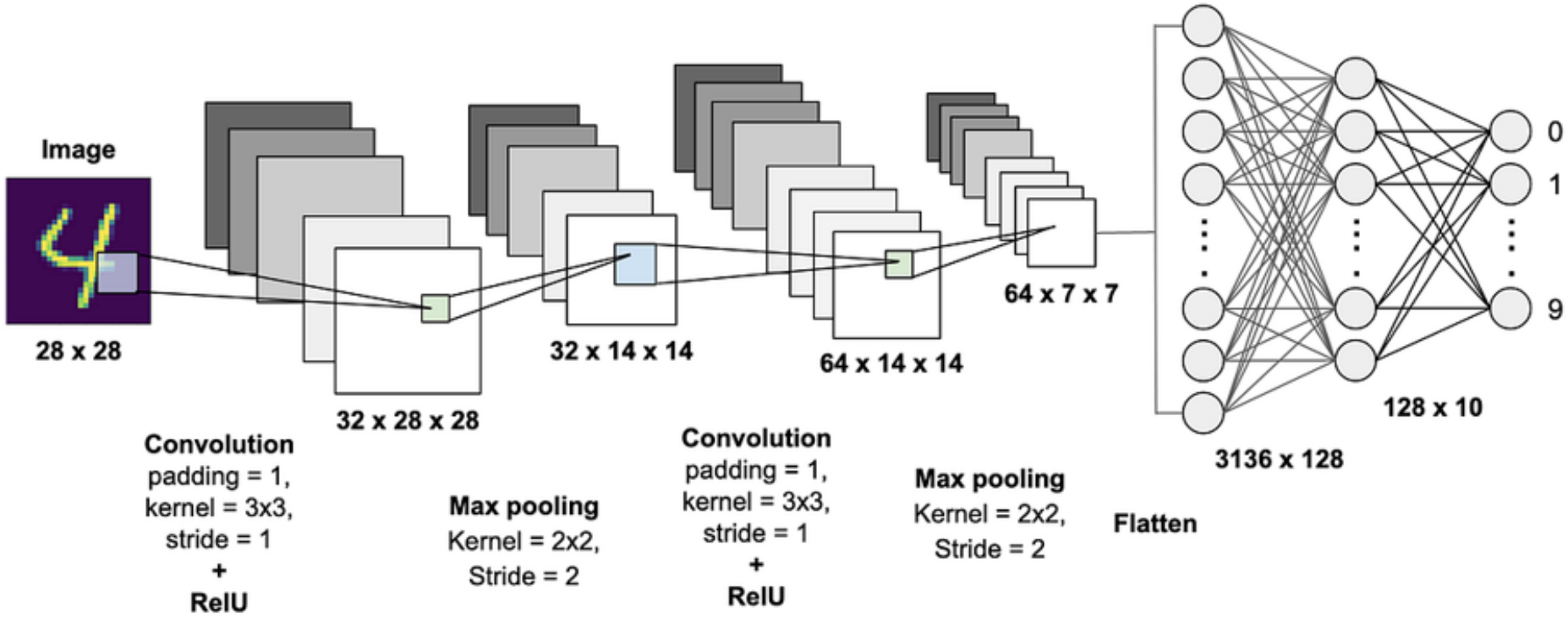


DOMAIN SHIFT

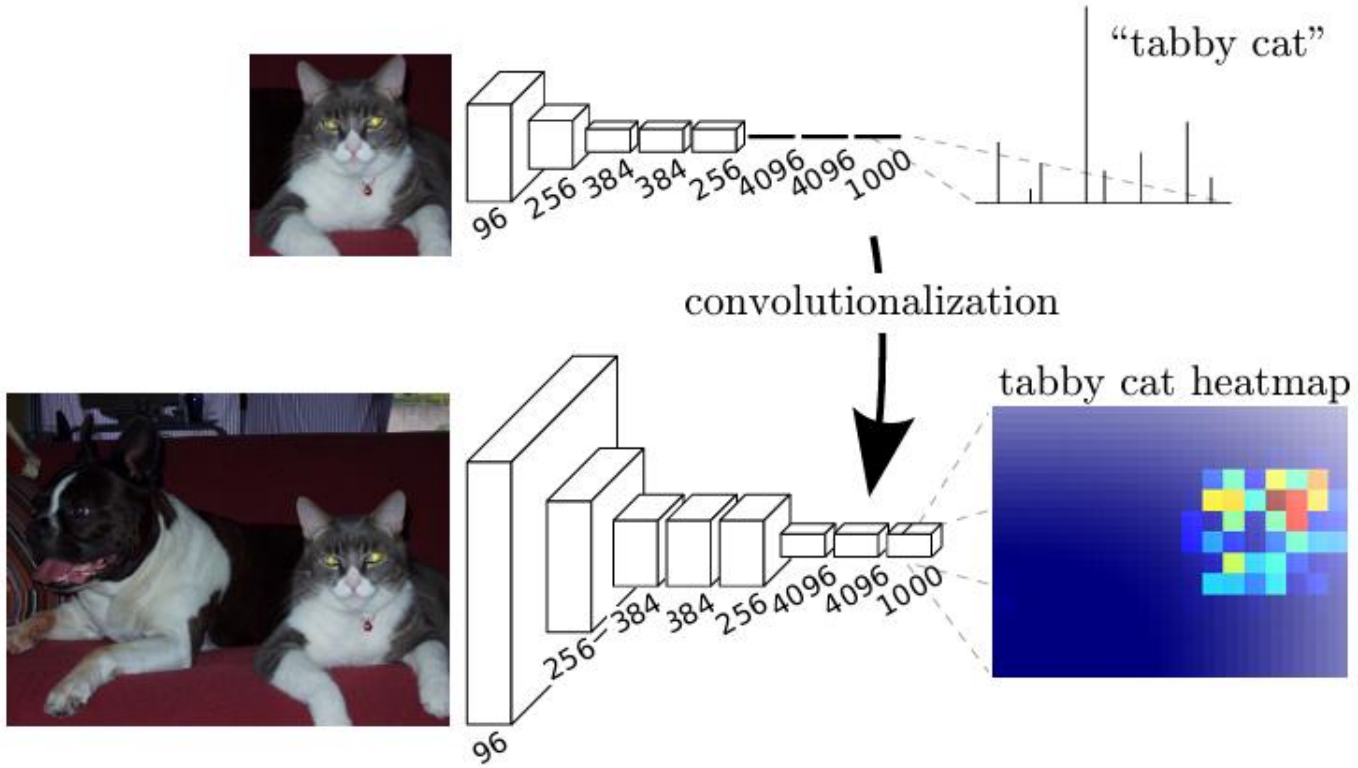




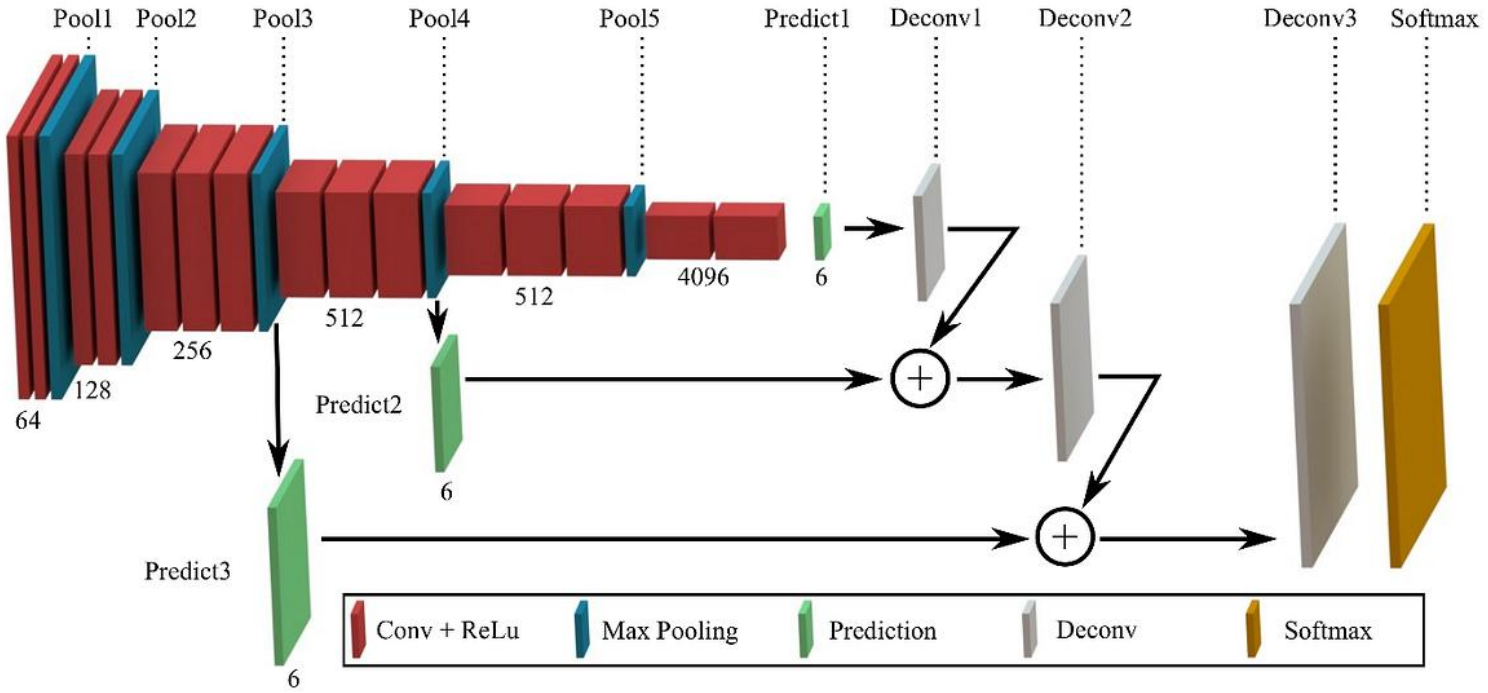
CONVOLUTIONAL NEURAL NETWORK

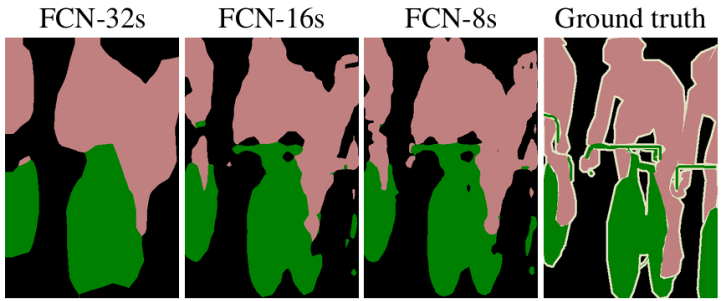


FULLY CONVOLUTIONAL NEURAL NETWORK

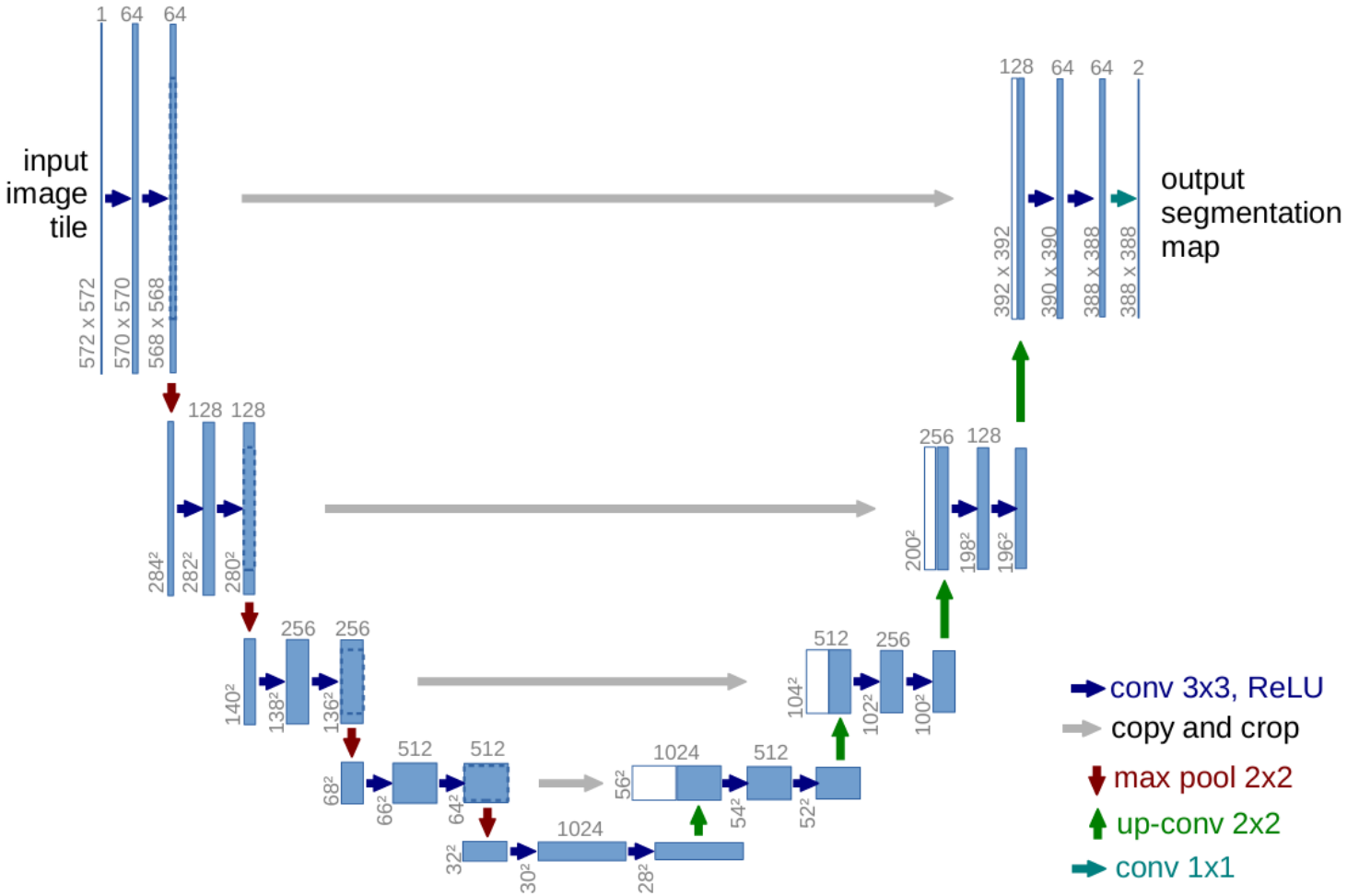


DECODER & SKIP CONNECTIONS

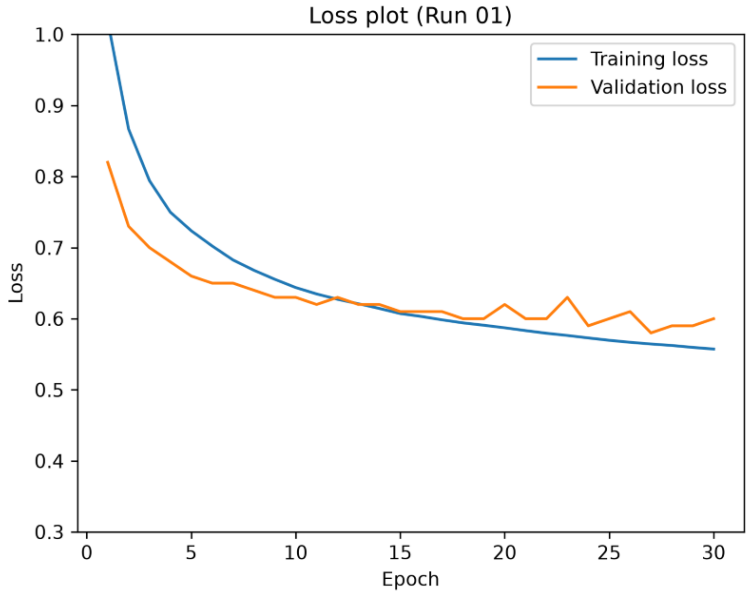


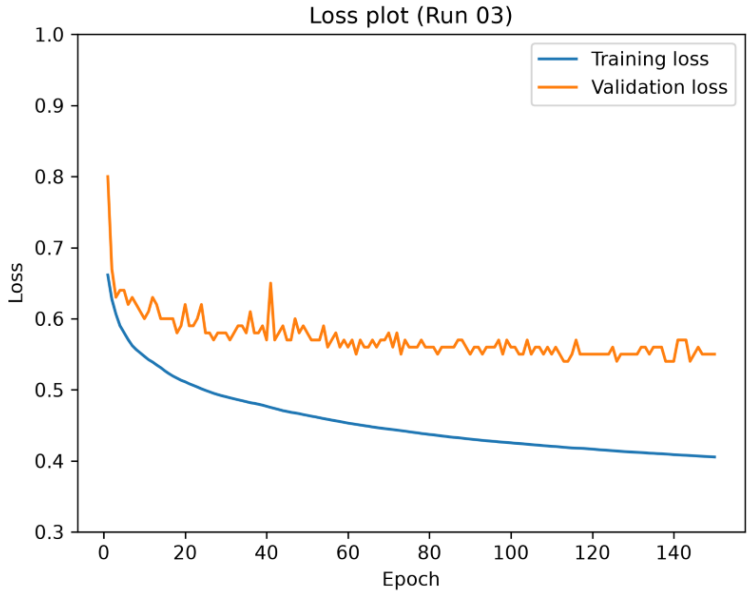


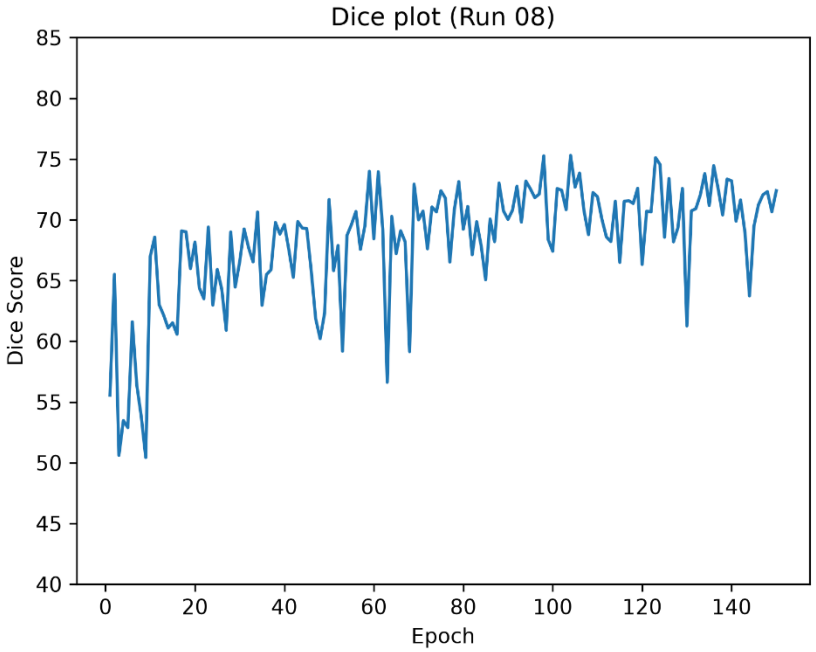
U-NET


input image tile
572 x 572
570 x 570
568 x 568
1 64 64
128 128
284²
282²
280²
256 256
140²
138²
136²
512 512
68²
66²
64²
1024
32²
30²
28²
1024
56²
512
54²
512
52²
512 256
104²
102²
100²
256 128
200²
198²
196²
128 64 64 2
392 x 392
390 x 390
388 x 388
388 x 388
output segmentation map
conv 3x3, ReLU
copy and crop
max pool 2x2
up-conv 2x2
conv 1x1

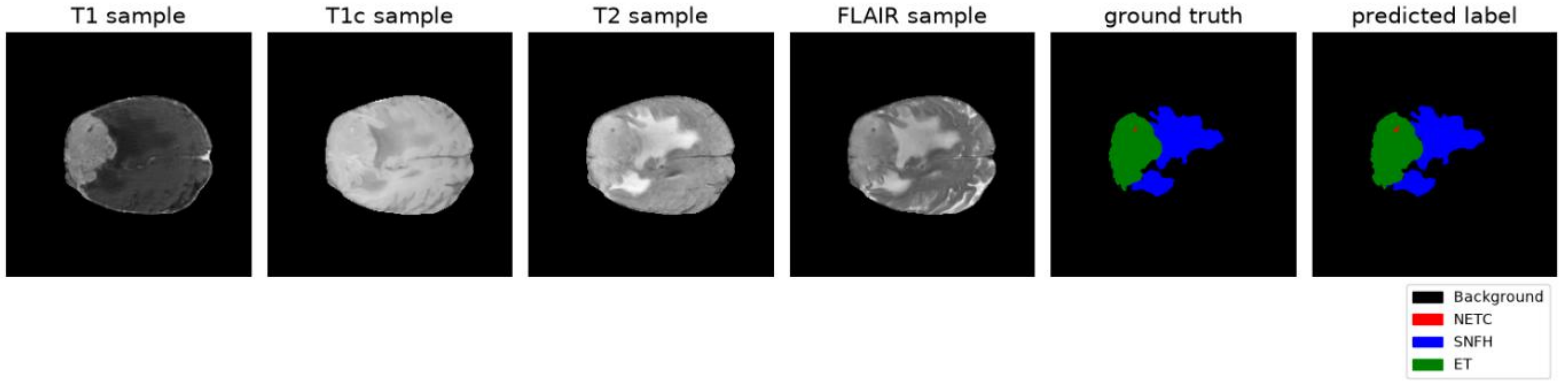
TRAINING BASE MODEL

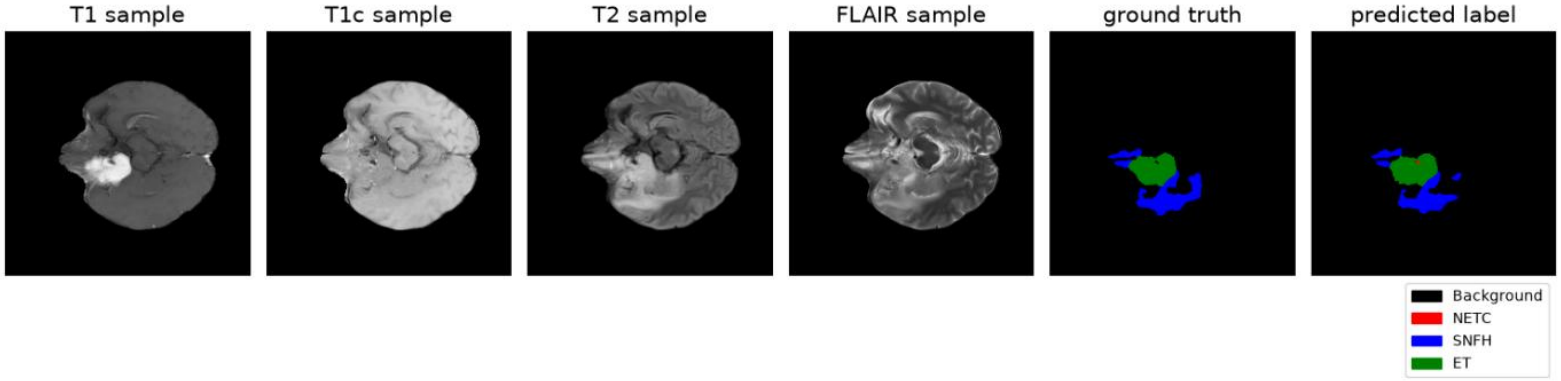


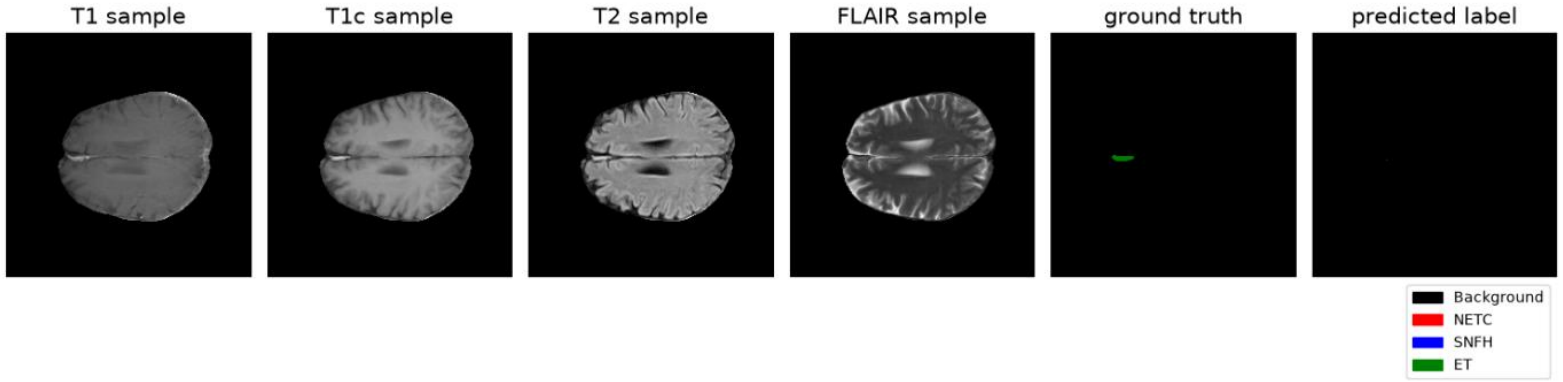




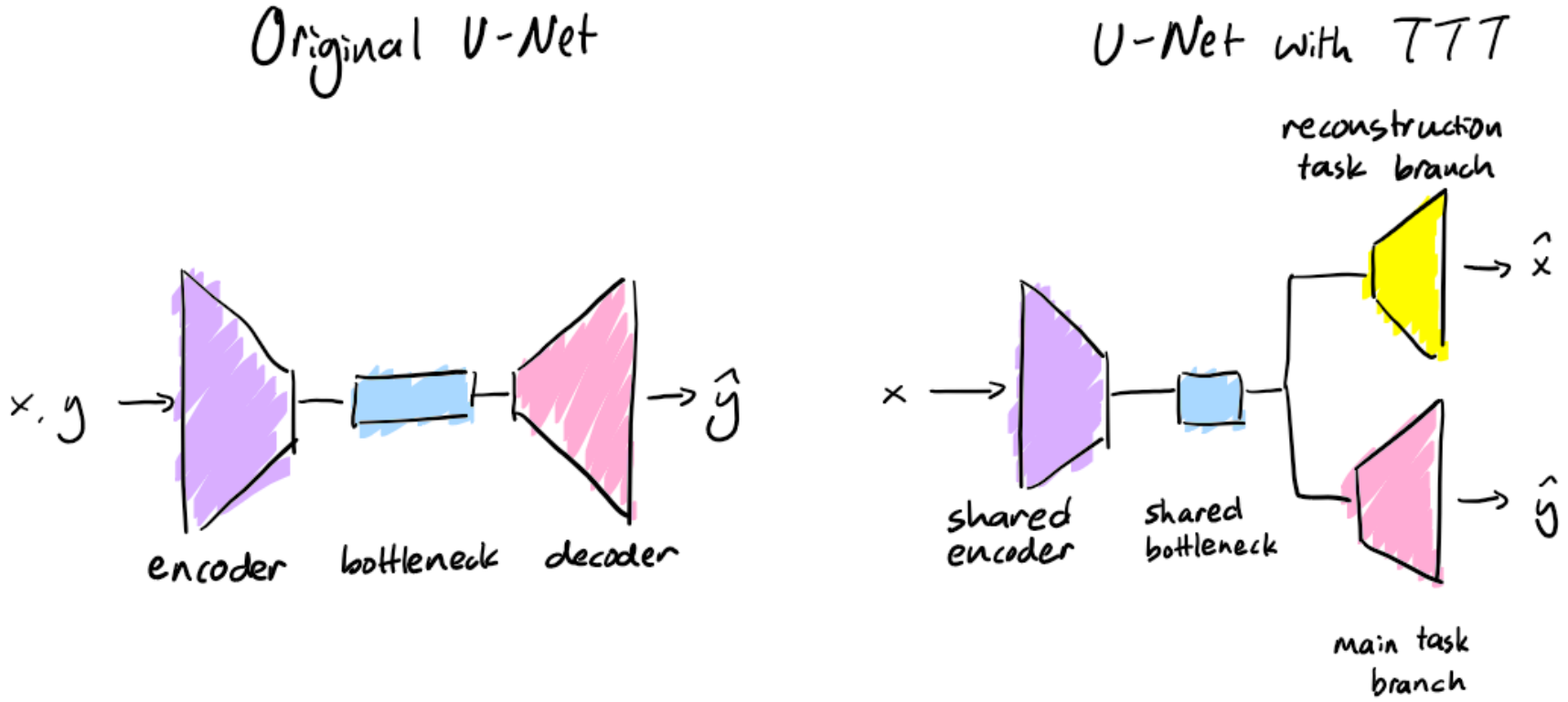
SEGMENTATION RESULTS



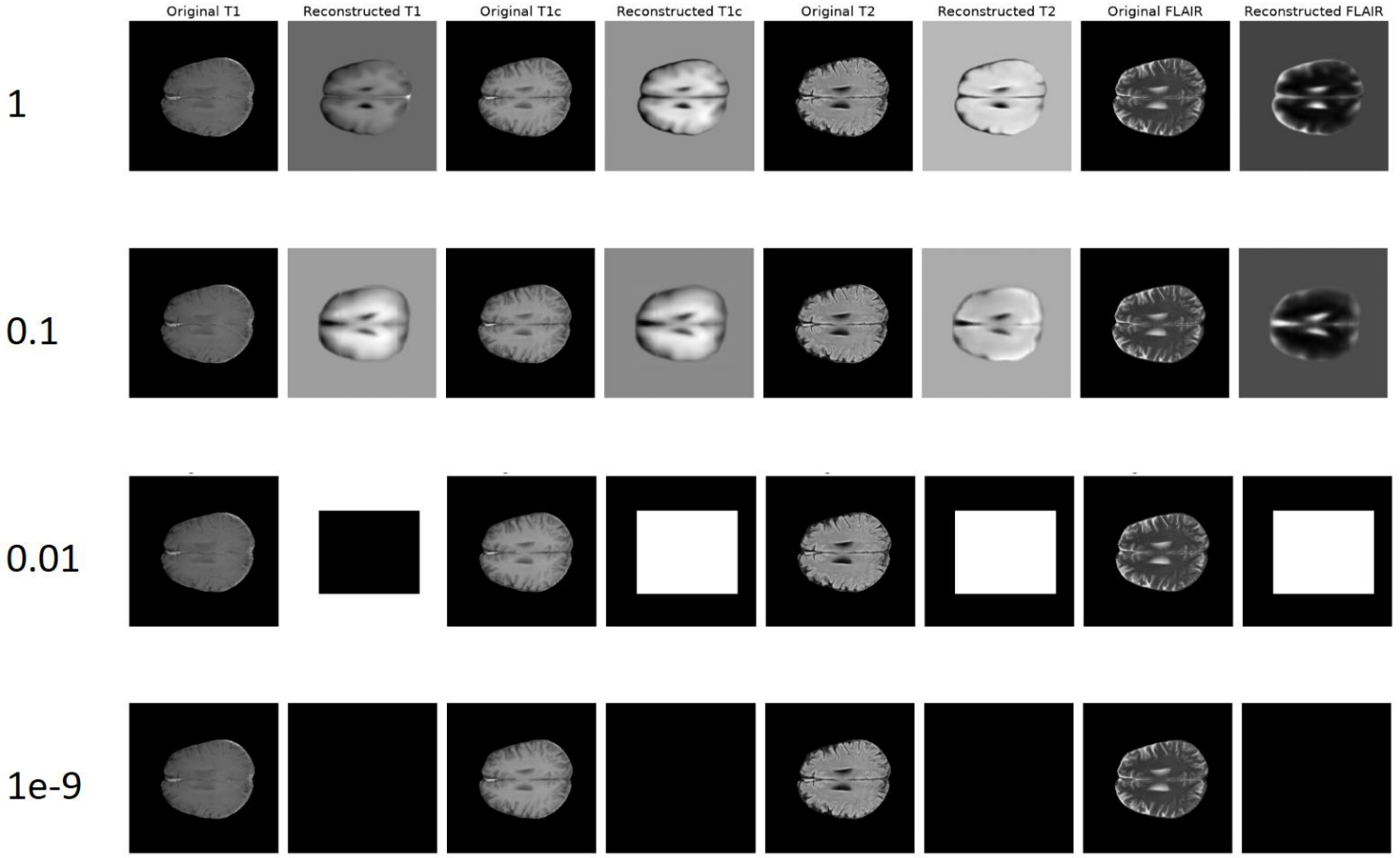




TEST-TIME TRAINING



RECONSTRUCTION EXAMPLES



MODEL PERFORMANCE

Before TTT:

	D1	D2	D3	Average
Meningioma	75.7	75.10	73.80	74.90
Glioma	37.30	45.15	67.50	50.00
Other Neoplasms	29.52	35.56	76.70	47.30

After TTT:

	D1	D2	D3	Average
Meningioma	77.40	65.30	77.04	73.24
Glioma	49.65	61.65	72.17	61.16
Other Neoplasms	34.79	52.56	75.24	54.20